What Catch Your Attention in SAR Images: Saliency Detection Based on Soft-Superpixel Lacunarity Cue

Fei Ma, *Member, IEEE*, Xuejiao Sun, Fan Zhang, *Senior Member, IEEE*, Yongsheng Zhou, *Member, IEEE*, and Heng-Chao Li, *Senior Member, IEEE*

***Abstract*— In existing superpixel-wise saliency detection algorithms, superpixel generation often is an isolated preprocessing step. The performance of saliency maps is determined by the accuracy of superpixels to a certain extent. However, it is still a challenge to develop a stable superpixel generation method. In this article, we attempt to incorporate the superpixel generation and saliency calculation steps into an end-to-end trainable deep network. First, we employ a recently proposed differentiable superpixel generation method to over-segment the synthetic aperture radar (SAR) images, which outputs the possibility that the pixels assigned to neighbor superpixels (soft superpixel). In saliency calculation part, as one of our main contributions, we propose a differentiable and computationally simple saliency model, i.e., lacunarity cue. It is inspired by the fact that generally the backscattering intensity of regions of interest (ROIs) in SAR images irregularly fluctuates, while the areas with consistent pixels are often ignored as the clusters. We improve the pixelwise box differential dimension algorithm to measure the irregularity of scattering points in a superpixel. The superpixel generation and saliency calculation can be implemented under a unified deep network. Hence, the shapes of the superpixels can be iteratively adjusted according to the saliency maps until the ROIs are correctly detected. Experiments on real SAR images with different sizes and scenes show that the saliency maps can effectively highlight the target areas, thus outperforming the state-of-the-art saliency detection models.**

***Index Terms*— Lacunarity saliency, saliency detection, synthetic aperture radar (SAR).**

I. Introduction

AS AN active microwave imaging sensor, synthetic aperture radar (SAR) can achieve all-day and all-weather observation, and has the ability to penetrate certain obstacles to observe targets hidden under vegetation. Therefore, it plays an important role in both civil and military fields [1]. Achieving fast and accurate automatic interpretation of SAR images is important in the application.

Among the SAR image target detection methods, constant false alarm rate (CFAR) [2] is the most common supervised detection algorithm at present. In the 1990s, Lincoln Laboratory researchers proposed the two-parameter CFAR detection method [3] and applied it to the SAR interpretation system. Since then, SAR target detection algorithms for different clutter statistics have been proposed; Gao et al. [4] proposed the adaptive fast CFAR algorithm based on automatic censoring. Qu et al. [5] proposed the dual-threshold fast CFAR detection, and Leng et al. [6] proposed a CFAR target detection algorithm based on 2-D Ostu segmentation to improve the detection performance of dense targets. The basic idea of these algorithms depends on difference between target and background distributions to obtain an appropriate threshold. However, in complex scenes, CFAR-based models cannot fit diverse clutter perfectly [7], [8]. Therefore, it is difficult to obtain excellent detection results.

With the intensive research on cognitive psychology [9] and neuroanatomy [10], saliency detection based on visual attention models [11] have occupied a place in the field of detection [12]. When processing images, the human vision system can quickly search, locate, and track the area of interest, while ignoring the background areas, so as to complete the accurate extraction of targets. This fast and efficient image interpretation benefits from the human visual attention mechanism, which is a selective mechanism that guides people to focus on more salient objects and ignore other regions in the scene [13]. Inspired by this mechanism, researchers introduced the concept of saliency detection into the image interpretation tasks by simulating the function of the visual system. The purpose of saliency detection is to estimate the probability of each region in a given image to attract people's attention and use the saliency values to represent this probability, which is characterized by grayscale, so as to screen out the areas of people's interest [14].

At present, saliency detection based on visual mechanisms has been widely used in optical images. The pioneering work is Itti model [15], which performs center-surround contrast processing based on three feature maps of multiscale intensity, color, and orientation to achieve effective saliency information extraction. This pioneering research has laid the foundation for follow-up work. Some scholars have successively proposed many classical models. For example, Cheng et al. [16] proposed a salient area detection algorithm based on global contrast. Hou and Zhang [17] and Achanta et al. [18] introduced frequency-domain information into detection and

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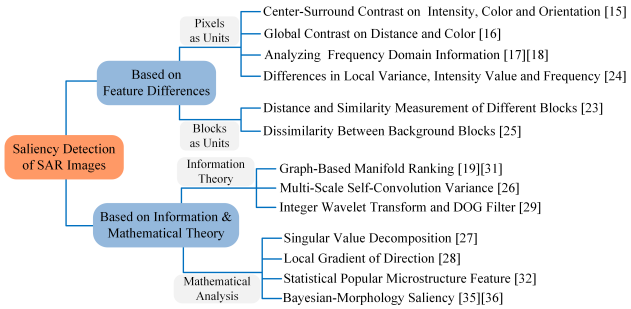

Fig. 1. Summary of saliency methods for SAR images.

proposed spectral residual (SR) and frequency-tuned (FT) algorithm, respectively, where salient regions were extracted by analyzing the frequency-domain information of images. Yang et al. [19] used graph-based manifold ranking (GMR) to rank the input images based on graph theory to detect the salient regions.

In addition to optical images, saliency detection is also used to simplify scene analysis and reduce computational complexity for SAR image interpretation [20]. However, compared with optical images, SAR has special imaging mechanism. Each pixel in the SAR images is largely determined by the intensity of scattered echoes after the ground object receives side-viewing radar irradiation [21]. At the same time, the existence of coherent speckle noise greatly increases the difficulty of SAR image interpretation. Therefore, traditional optical image saliency detection algorithm cannot be directly applied to SAR images.

In recent years, many researchers have proposed some applicable saliency detection algorithms for SAR images. Yu et al. [22] first applied the frequency domain-based method to SAR images in simple scenes and achieved good results, which demonstrated the feasibility of the saliency detection model in SAR images and opened the precedent of research in this field. On this basis, many algorithms have been proposed for SAR images. Current saliency detection models for SAR images can be roughly divided into two categories. One is the traditional algorithms based on the scarcity of target features, and the other is the models which combine saliency map construction with neural networks.

Most of the algorithms based on the scarcity of target features are the improvements of the optical image methods based on the characteristics of SAR images. These methods can be roughly divided into two categories. As shown in Fig. 1, in the first category, the saliency value of a pixel is constructed by difference between its intensity, texture features, and that of its globally or local neighbor pixels. For example, Huang et al. [23] started from the distance measurement of different image blocks and constructed a saliency map by accurately measuring the similarity between different image blocks. Lai et al. [24] obtained differences in features such as local variance, intensity value, and frequency from SAR images and merged them to form a saliency map. Ni et al. [25] used the dissimilarity between background image blocks and adaptively selected image blocks to construct a saliency map. These algorithms verified the effectiveness on simple-scene SAR images. In the second category, many scholars proposed to construct saliency maps using information theory or other mathematical analysis algorithms. For instance, Wang et al. [26] completed the saliency detection by calculating multiscale self-convolution variance. In order to improve the detection capability of the Itti model for weak targets, Liu et al. [27] proposed a saliency detection algorithm for SAR images using singular value decomposition. Dou et al. [28] constructed a saliency map based on the local gradient of the direction to locate the target location. Zhang and Liu [29] used integer wavelet transform and difference of Gaussian (DOG) filter to obtain feature maps of azimuth and intensity. Li et al. [30] constructed saliency maps by fusing the feature maps of standard deviation, peak value, and background at the superpixel level, and extracted the target region using global threshold. Later, Wang et al. [31] used Markov random field and fuzzy mean to construct a saliency map. Recently, Yang et al. [32] proposed a statistical popular microstructure feature analysis method based on curvature to construct the saliency maps for target detection. In recent years, Bayesian model [33] is also widely applied to construct the saliency maps. In order to remove the strong clutter in nonuniform background, inspired by scale selection [34], Wang et al. [35] proposed a Bayesian-morphology saliency map construction algorithm. It introduced the size prior information of the targets to effectively suppressing clutter. Wang et al. [36] integrated texture saliency maps and Bayesian saliency maps to highlight the target areas. Then, the fine target contours are acquired through active contour models.

The above algorithms comprehensively can achieve effective detection in simple scenes, but they are still far from application in real world. First, these algorithms are directly migrated from optical images without fully considering the unique imaging mechanism and image characteristics of SAR images. At the same time, the pixel-by-pixel processing has high calculation complexity, especially considering that SAR images have large scenes. In addition, Bayesian-based methods also require the prior and likelihood maps, which provide rough foreground and background regions for the construction of Bayesian saliency maps, which makes the detection steps tedious. Based on the above analysis of existing algorithms, we believe that the saliency detection algorithm for SAR images should be fully consider the unique characteristics of SAR images.

To address the above problems, we propose a soft superpixel-wise saliency method based on the difference between the fluctuations of scattering points in the regions of interest (ROIs) and natural terrains. We find that generally the areas where the backscattering intensity irregularly fluctuates in a wide range easily attract human's attention. On the contrary, the areas with consistent pixel values are often ignored as the background. Taking vehicles as example, they generally have unregular geometric shapes, and different parts have various scattering mechanisms. The middle parts of the vehicles mostly generate the dihedral or trihedral scattering, thus resulting in high backscattering intensity, while some parts present the primary scattered waves and the intensity

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is relatively weak. Hence, they are likely to attract human's attention.

In addition, in the previous superpixel-wise saliency algorithms, superpixel generation is always separated from the saliency calculation. Once superpixels are generated, their shapes cannot be changed. In this case, the performance of the saliency is determined by the accuracy of superpixels to a certain extent. To address this, a soft superpixel generation method [37] is introduced to construct accurate saliency maps. Instead of the hard association between pixels and superpixels, this method utilizes the probabilities to represent the relationship between superpixels and pixels [38], making the superpixel generation differentiable. Hence, it can be implemented using a deep fully convolutional network and the shapes of the superpixels can be iteratively adjusted according to the saliency results in the training process.

The method mainly includes the following two innovations.

1) Inspired by the fact that highly variable areas in SAR images easily attract people's attention while the stable areas are often ignored as the background, we construct a computationally simple and differentiable superpixel-wise lacunarity saliency. We improve the classical pixelwise box differential dimension algorithm to measure the irregularity of scattering points in a superpixel. It can distinguish the background clutters from ROIs based on the difference in the fluctuations of scattering points between the ROIs and background. Before this, rare superpixel-wise saliency methods could be implemented using deep networks.
2) Based on the above work about differentiable superpixel generation, we construct an end-to-end superpixel-wise saliency prediction network combining superpixel generation and lacunarity saliency calculation. Superpixels' shapes are iteratively adjusted according to the saliency maps until the parameters capture the globally optimal solution.

The rest of this article is organized as follows. Section II introduces the motivation and classical pixelwise lacunarity saliency. Section III describes the proposed saliency method. Section IV analyzes the experimental results on real SAR images. Finally, Section V concludes this article.

II. What Catch Your Attention in SAR Images: Motivations of Lacunarity Feature

A. Motivation of Lacunarity Saliency Cue

In single-channel SAR images, the ROIs generally refer to vehicles, ships, or other man-made targets. We find that the scattering mechanisms of different parts of the targets are quite different. As shown in Fig. 2(a), the pixel values of building areas fluctuate greatly, that is, strong scattering points are irregularly distributed in these areas. Hence, a large gray value mutation occurs in the target areas, forming a large size of the "gap." In general, these areas in SAR images easily attract human's attention. In contrast, the natural terrains have consistent backscattering coefficients, so that the pixel values of these areas have less fluctuation [Fig. 2(b)]. These areas are often ignored as the background. Thus, we propose a soft-superpixel-wise saliency cue for ROI detection based on the irregularity of the scattering points, i.e., lacunarity feature.

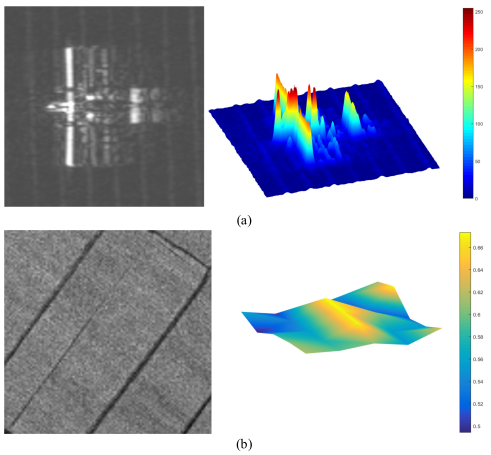

Fig. 2. Comparison of backscattering coefficients of ROIs and background: (a) backscattering intensities of ROIs dramatically fluctuates, which easily attract human's attention and (b) backscattering intensities of background change slowly, which often gets neglected.

B. Pixelwise Lacunarity Feature Based on Differential Box Dimension

Differential box dimension algorithm [39] is a classical method to measure the irregularity of an image. Based on the idea of differential box dimension, Li et al. [30] proposed the pixelwise lacunarity saliency for target detection in SAR images. This section will describe the details of this kind of saliency.

Let $\mathbf{I}$ represents the input image of size $W \times H$. The image is first mapped into a 3-D plane, where x and y coordinates represent the length and width of $\mathbf{I}$, respectively, and z-coordinate denotes the intensity values of $\mathbf{I}$. The xy plane is partitioned into nonoverlapping grids of size $s \times s$ pixels, where s is an integer and s varies from 2 to S. If the input image $\mathbf{I}$ is a square of size $W \times W$, S is $W/2$. Boxes of size $s \times s \times s$ are required to be placed one after another on each grid to represent gray-level variations of that grid. Let m and n denote the serial number of boxes containing the maximum and minimum gray levels on a grid, respectively. The fractal dimension of this grid is defined as $|m - n| + 1$. Let N_s be the summation of fractal dimensions of all grids at scale s. N_s indicates the total number of required boxes at scale s to cover the rough intensity surface of an image. Let $y_s = \log N_s$ and $x_s = \log(1/s)$. Therefore, there are a series of y_s and x_s under different s. Finally, (y_s, x_s) is fit by linear least squares and the fit slope is defined as the fractal dimension of this image [39]

$$D = \frac{S \cdot \sum_{s=2}^{S} x_s y_s - \sum_{s=2}^{S} x_s \sum_{s=2}^{S} y_s}{S \cdot \sum_{s=2}^{S} x_s^2 - \left(\sum_{s=2}^{S} x_s\right)^2}. \tag{1}$$

A large difference box dimension indicates that the image has strong texture and edge features. However, the differential box dimension can only measure the irregularity of the whole image and cannot be used to calculate the irregularity of a

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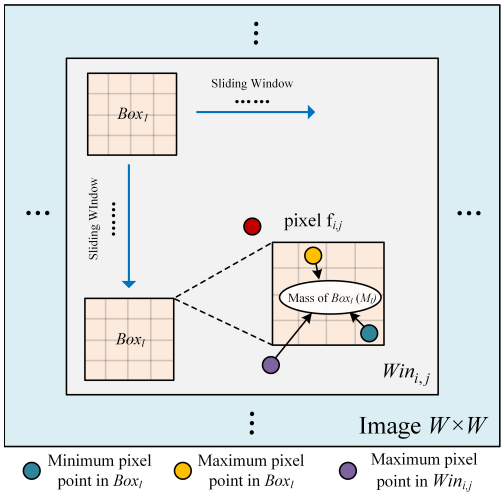

Fig. 3. Pixelwise lacunarity feature of a pixel based on differential box dimension.

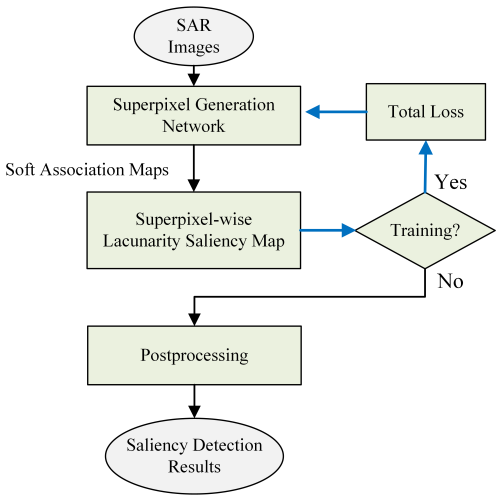

Fig. 4. Framework of proposed saliency detection method.

certain pixel or region. For this problem, Wang et al. [40] proposed pixelwise SAR image lacunarity feature based on the idea of differential box dimension algorithm. A sliding window is used to replace the grid with fixed position. The specific process is shown in Fig. 3. Assuming the input's size is $W \times H$, for a pixel $f_{i,j}$, first we select a window of size $R \times R$ centered on $f_{i,j}$, denoted as $\text{Win}_{i,j}$. Then, a small box with a size of $L \times L$ slides in this window, and the lth box is recorded as Box_l. Wang et al. [40] defined the mass of the Box_l as

$$M_l = \frac{c \times [\max(\text{Box}_l) - \min(\text{Box}_l)]}{\max\left(\text{Win}_{i,j}\right) \times L} \tag{2}$$

where c is the quantization coefficient. $\max(\text{Box}_l)$ and $\min(\text{Box}_l)$ are the maximum and minimum pixels in the box Box_l. $\max(\text{Win}_{i,j})$ is the maximum pixel in $\text{Win}_{i,j}$. It can be seen that boxes in different positions of the same window have different masses. The mass of each box M_l can reflect the irregularity of backscattering intensities inside the box. When $M_l = 0$, it means that amplitudes of all pixels inside the box are equal, which is a strictly uniform region. Small M_l indicates the pixels inside box have low fluctuation. When the pixels amplitude inside the box fluctuates violently, M_l is high.

Making the box Box_l slides in the window, a series of masses can be obtained according to (2). Then, the lacunarity of the pixel $f_{i,j}$ is defined as

$$\Lambda_{i,j} = \frac{\sum_{M_l} M_l^2 \times P(M_l)}{\left[\sum_{M_l} M_l \times P(M_l)\right]^2} \tag{3}$$

and

$$P(M_l) = \frac{\text{num}(M_l)}{(R - L + 1)^2} \tag{4}$$

where $\text{num}(M_l)$ is the number of boxes with mass M_l in this window and $(R - L + 1)^2$ is the total number of boxes in this window when the box moves in one-pixel step. $P(M_l)$ denotes the statistical probability of mass M_l.

It can be seen that the lacunarity of pixel $\Lambda_{i,j}$ is the ratio of the second-order origin moment to the squared first-order moment of a series of box masses within the window. It can describe the fluctuation of the box mass M_l within the window $\text{Win}_{i,j}$, reflecting the irregularity of pixels in $\text{Win}_{i,j}$. The small $\Lambda_{i,j}$ indicates that the statistical distribution of the mass M_l in the window area is relatively uniform. When there is a larger fluctuation, the $\Lambda_{i,j}$ will be larger. Taking advantage of the difference in scattering irregularity between targets such as vehicles and the natural terrains, the background clutters can be removed, while the target areas with larger lacunarity will be retained.

III. Proposed Saliency Detection Method

A pixel-by-pixel sliding window used in [40] has heavy computational burden when process the large-scene SAR images. To solve this problem, we propose the superpixel-wise lacunarity saliency, as illustrated in Fig. 4.

First, the input SAR image is over-segmented by a deep superpixel generation network proposed in our previous achievement [41]. It is an encoder–decoder network, where encoder extracts the features and decoder outputs the likelihood that each pixel belongs to different superpixels, i.e., association map. Then, we propose the superpixel-wise lacunarity saliency based on the association map. In the training process, the network can iteratively optimize the shapes of superpixels according to the saliency maps. However, there are still some man-made objects or natural terrains that are given high saliency in the saliency maps, because of the large mutations of their pixel values. The following postprocessing (PPR) is introduced to effectively highlight the target areas and suppress the clutters using the prior information about the targets, thus obtaining the final detection results with fewer false alarms and higher detection rate.

A. Soft Superpixel Generation Network

Due to speckle noise and complex geometric distortions within SAR images, there are always some superpixels that cannot adhere the real boundaries well. For example, a part

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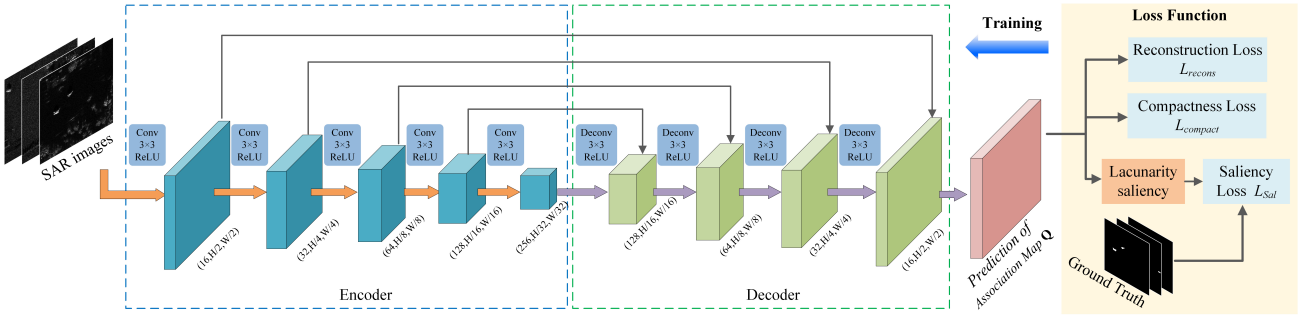

Fig. 5. Structure and training process of superpixel generation network.

of ROIs and background regions are clustered into the same superpixels. These worry superpixels will directly lead to errors in the subsequent saliency calculation. We hope that the shapes of superpixels can be updated based on the saliency maps, so as to optimize the saliency results. In this article, we introduce a soft superpixel generation method proposed in our previous work [41]. This method is differentiable and can be realized by an encoder–decoder deep convolutional network. Its structure and training process are shown in Fig. 5. Let $\mathbf{I}$ represents an input SAR image of size $W \times H$. The encoder extracts the high-level feature maps from SAR images, which contains five convolution layers with 16, 32, 64, 128, and 256 convolution kernels of size 3×3, respectively. Correspondingly, the decoder is composed of four convolution layers with 128, 64, 32, and 16 convolution kernels. It projects the feature maps to a soft association map, i.e., $\mathbf{Q} \propto \mathbb{R}^{W\times H\times M}$, where entry $q^m_{i,j}$ denotes the probability of assigning pixel $f_{i,j}$ to the mth superpixel. M represents the number of superpixels in image I.

As described in [40], two kinds of unsupervised loss are calculated based on soft association map $\mathbf{Q}$, i.e., reconstruction loss and compactness loss. To be specific, reconstruction loss encourages the superpixel prediction network to group pixels with similar property as far as possible. Let $a_{i,j}$ and $p_{i,j} \propto \mathbb{R}^{1\times 2}$ represent the intensity and coordinates of pixel $f_{i,j}$. We can obtain the intensity and location of mth superpixel's center, expressed as $\hat{a}_m$ and $\hat{p}_m$

$$\hat{a}_m = \frac{\sum_{(i,j)\in \mathrm{I}} a_{i,j} \cdot q^m_{i,j}}{\sum_{(i,j)\in \mathrm{I}} q^m_{i,j}}, \quad \hat{p}_m = \frac{\sum_{(i,j)\in \mathrm{I}} p_{i,j} \cdot q^m_{i,j}}{\sum_{(i,j)\in \mathrm{I}} q^m_{i,j}} \tag{5}$$

where $\sum_{(i,j)\in \mathrm{I}} q^m_{i,j}$ is the normalization constant. Mathematically, $\hat{a}_m$ and $\hat{p}_m$ are the weighted mean of intensity values and locations of all pixels assigned to it.

Then, given all superpixel centres' properties, we can reconstruct the intensity and location of each pixel as follows:

$$\hat{a}_{i,j} = \frac{\sum_{m=1}^{M} \left(\hat{a}_m \cdot q^m_{i,j}\right)}{\sum_{m=1}^{M} q^m_{i,j}}, \hat{p}_{i,j} = \frac{\sum_{m=1}^{M} \left(\hat{p}_m \cdot q^m_{i,j}\right)}{\sum_{m=1}^{M} q^m_{i,j}} \tag{6}$$

where $\hat{a}_{i,j}$ and $\hat{p}_{i,j}$ are, respectively, the reconstructed intensity and position of pixel $f_{i,j}$. $\sum_{m=1}^{M} q^m_{i,j}$ is the normalization constant, meaning that the total probability that pixel $f_{i,j}$ belongs to all superpixels is one. The physical meaning of (6) is that the reconstructed features of each pixel are the weighted average of the features of all superpixels related to it.

In an ideal association map, the neighbor and similar pixels will be merged into the same superpixels. The initial and property of each pixel will be very close to its reconstructed property. From this, the reconstruction loss $L^{\mathrm{u}}_{\mathrm{recons}}$ can be defined as the difference between original and reconstructed properties of all pixels

$$L^{\mathrm{u}}_{\mathrm{recons}} = \sum_{i=0,j=0}^{i=H,j=W} \left[\mathcal{L}\left(a_{i,j}, \hat{a}_{i,j}\right) + \mathcal{L}\left(p_{i,j}, \hat{p}_{i,j}\right)\right] \tag{7}$$

where $\mathcal{L}(\cdot)$ is the cross-entropy function. This loss encourages the trained model to group pixels with similar property.

Like conventional superpixel generation method, we also introduce the compactness loss to encourage the superpixels to be spatially compact, which is defined as

$$L_{\mathrm{compact}} = \frac{\sum_{m=1}^{M} \sum_{(i,j)\in \mathrm{I}} \|\hat{p}_m - p_{i,j}\|_2}{H \cdot W \cdot M} \tag{8}$$

where $\|\cdot\|_2$ is the L2-norm. The compactness loss can minimize the mean spatial distance between the position of superpixel center and all pixels inside it. Further details about soft-superpixel generation can be found in [41].

In addition to the above two loss functions, we also propose saliency loss in this article. It is calculated based on the difference between the superpixel-wise lacunarity saliency map (LSM) and the ground truth (GT). It enables the superpixel generation network to update the shape of the superpixels based on the results of lacunarity saliency calculation. In Section III-B, we will elaborate on the lacunarity saliency and the corresponding loss.

B. Superpixelwise Lacunarity Saliency

As shown in Fig. 6, we divide the superpixel-wise lacunarity saliency into two parts. First, we find that if the distribution of scattering intensity within a superpixel is irregular, that is, there are multiple types of backscattering points inside the superpixel, the superpixel is likely to be located in the target regions. It should be given high saliency. Another part of the saliency of a superpixel comes from its contrast with the surrounding superpixels. When a superpixel's lacunarity is quite different with that of other neighbor superpixels, it is likely that the superpixel easily attracts human's attention, so that the superpixel should be given higher saliency.

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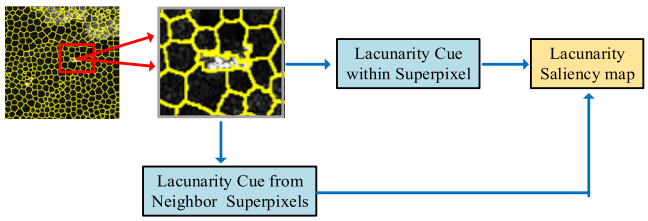

Fig. 6. Construction method of LSM.

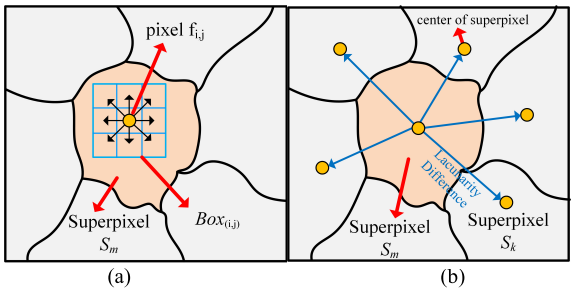

Fig. 7. Construction of LSM: (a) lacunarity saliency in superpixel and (b) lacunarity saliency from the surrounding superpixels.

For the mth superpixel S_m, its lacunarity saliency is calculated as shown in Fig. 7(a). $f_{i,j}$ is a pixel in superpixel S_m and a square region of size $L \times L$ centered on that pixel is selected as its box, i.e., $\mathrm{Box}_{(i,j)}$. The mass of the box can be calculated according to (2), denoted as $M_{(i,j)}$. By selecting different pixels in this superpixel in turn, a series boxes and their masses belonging to superpixel S_m are obtained. Based on (3), the lacunarity saliency value of this superpixel is defined as

$$\mathrm{Sal}_{\mathrm{in}}^{m} = \frac{\sum_{(i,j)\in \mathrm{I}} \left(M_{i,j} q_{i,j}^{m}\right)^2 \times \rho^{m}\left(M_{(i,j)}\right)}{\left[\sum_{(i,j)\in \mathrm{I}} \left(M_{i,j} q_{i,j}^{m}\right) \times \rho^{m}\left(M_{(i,j)}\right)\right]^2} \tag{9}$$

and

$$\rho^{m}\left(M_{(i,j)}\right) = \frac{\sum_{(i,j)\in \mathrm{I}} \left[q_{i,j}^{m} \middle| M_{(i,j)}\right]}{\sum_{(i,j)\in \mathrm{I}}^{q_{i,j}^{m}}} \tag{10}$$

where $\rho^{m}(M_{(i,j)})$ represents the proportion of boxes whose mass is $M_{(i,j)}$ among all boxes belonging to the superpixel S_m, i.e., the statistical probability of mass $M_{(i,j)}$. $q_{i,j}^{m}$ denotes the probability of assigning pixel $f_{i,j}$ to the superpixel S_m. It should be noted that because pixel $f_{i,j}$ is assigned to the superpixel S_m with a probability $q_{i,j}^{m}$, the mass of the box centered on this pixel, i.e., $M_{(i,j)}$, also needs to be multiplied by the probability $q_{i,j}^{m}$ in (9). $\sum_{(i,j)\in \mathrm{I}}^{q_{i,j}^{m}}$ in (10) represents the number of pixels assigned in the superpixel S_m.

It can be seen that the larger variation of the masses of the boxes within the superpixel indicates that the distribution of the backscattering coefficients in the region is more irregular. It is more likely that the region belongs to the target region, thus giving higher saliency.

Moreover, if the lacunarity saliency of the superpixels is noticeably different with its neighbor superpixels, there is a high likelihood that a superpixel locates in the target areas and that its neighboring superpxies are in the backgrounds. It should be given high saliency. Hence, the difference in lacunarity saliency cues of adjacent superpixels also needs to be considered during calculating the lacunarity saliency. As shown in Fig. 7(b), for superpixel S_m, the saliency value from neighbor superpixels is defined as

$$\mathrm{Sal}_{\mathrm{out}}^{m} = \sum_{S_m \neq S_k} \omega_{\mathrm{size}}(S_k) \cdot \omega_{\mathrm{spatial}}(S_k, S_m) \cdot D_{\mathrm{La}}(S_k, S_m) \tag{11}$$

where $\omega_{\mathrm{size}(S_k)}$ is the size weight coefficient of S_k, expressed by the ratio of the areas of two superpixel S_m and S_k, that is,

$$\omega_{\mathrm{size}(S_k)} = \sum\nolimits_{(i,j)\in \mathrm{I}} q_{i,j}^{k} \Big/ \sum\nolimits_{(i,j)\in \mathrm{I}} q_{i,j}^{m} \tag{12}$$

In (11), $\omega_{\mathrm{spatial}}(S_k, S_m)$ is the spatial distance weight coefficient, which is expressed by the spatial Euclidean distance function between two superpixel regions

$$\omega_{\mathrm{spatial}}(S_k, S_m) = \exp\left(\left[-D_{\mathrm{spatial}}(\hat{p}_k, \hat{p}_{\boldsymbol{m}})\right]\right) \tag{13}$$

where $\hat{p}_k$ and $\hat{p}_{\boldsymbol{m}}$ are the coordinates of centers of S_m and S_k. $D_{\mathrm{spatial}}(\hat{p}_k, \hat{p}_{\boldsymbol{m}})$ is the Euclidean distance between $\hat{p}_k$ and $\hat{p}_{\boldsymbol{m}}$. It can be seen that the farther the superpixel S_k is from S_m, the smaller the weight coefficient $\omega_{\mathrm{spatial}}(S_k, S_m)$, and the smaller the influence of the superpixel S_k on the saliency of S_m. In (11), $D_{\mathrm{La}}(S_k, S_m)$ is the lacunarity contrast between the superpixels S_k and S_m, which is also calculated by the Euclidean distance to characterize the difference in the lacunarity between two superpixels.

Combining the above two parts, the lacunarity saliency value of pixel $f_{i,j}$ is defined as

$$\mathrm{Sal}_{(i,j)} = \frac{\sum_{m=1}^{M} \left(\mathrm{Sal}_{\mathrm{in}}^{m} \cdot \mathrm{Sal}_{\mathrm{out}}^{m} \cdot q_{i,j}^{m}\right)}{\sum_{m=1}^{M} q_{i,j}^{m}} \tag{14}$$

where $\sum_{m=1}^{M} q_{i,j}^{m}$ is the normalization constant, meaning that the total probability that pixel $f_{i,j}$ belongs to all superpixels is one. The physical meaning of (14) is that the lacunarity saliency value of each pixel is the weighted average of the saliency values of all superpixels assigned to it. It can be seen that when the pixel is more likely to belong to the certain superpixel, the saliency value of this pixel is more likely to be determined by the superpixel which it belongs to. From the saliency of the above two parts, the method fully considers the lacunarity within the superpixel and other surrounding superpixels. The regression to pixel level saliency makes the results more accurate compared with the same superpixel having uniform saliency.

C. End-to-End Training Process

In order to make the superpixels can adhere as much as possible to the boundaries of targets of interest, we design the lacunarity saliency loss to train the superpixel generation network. Assuming the saliency label of pixel $f_{i,j}$ is $\Phi_{i,j}$, the corresponding saliency loss is defined as

$$L_{\mathrm{Sal}} = \sum_{i=0,j=0}^{i=H,j=W} \mathcal{L}\left(\mathrm{Sal}_{(i,j)}, \Phi_{i,j}\right) \tag{15}$$

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TABLE I
TRAINING STEPS OF THE PROPOSED METHOD

Input: Unlabeled data $\mathbf{I}_{\mathrm{U}} \in \mathbb{R}^{W\times H}$ and labeled data $\mathbf{I}_{\mathrm{L}} \in \mathbb{R}^{W\times H}$
Output: Superpixel-wise lacunarity saliency maps $Sal_{(i,j)}$

For Number of training iterations **do**
  For an image $\mathbf{I} \in \{\mathbf{I}_{\mathrm{U}}, \mathbf{I}_{\mathrm{L}}\}$
    1. Decoder predicting soft association map $\mathbf{Q}$ of this input image;
    2. Obtaining the intensities and locations of all superpixels' centers;
    3. Reconstructing the intensity and position of each pixel $\mathrm{f}_{i,j}$ as (6);
    4. Calculating two kinds of unsupervised loss function L_{sp} and $L_{compact}$ as (7) and (8);
    5. Calculating superpixel-wise lacunarity saliency $Sal_{(i,j)}$ as (14);
    If $\mathbf{I} \in \{\mathbf{I}_{\mathrm{L}}\}$
      6. Calculating the saliency loss L_{Sal} as (15);
    End if
  End for
  7. Update the parameters of superpixel generation networks based on L_{sp}, $L_{compact}$ and L_{Sal}.
End for

where $\mathcal{L}(\cdot)$ is the cross-entropy function. If the superpixels poorly adhering the real boundaries of targets, that is, the target pixels and adjacent background pixels are clustered into the same superpixel, the saliency results will have a large difference with GT. The value of significance loss will become larger, thus updating the target parameters faster.

In summary, based on (7) and (8), the loss of this end-to-end saliency method can be written as

$$L_{\mathrm{total}} = L^{\mathrm{u}}_{\mathrm{recons}} + \alpha \cdot L_{\mathrm{compact}} + \beta \cdot L_{\mathrm{Sal}} \tag{16}$$

where α and β are the weights of compactness loss and saliency loss to balance the merging loss term. The first item, reconstruction loss $L^{\mathrm{u}}_{\mathrm{recons}}$, encourages the trained model to group pixels with similar property. The second item, compactness loss L_{compact}, enforces the superpixels to be spatially compact. The third item, saliency loss L_{Sal}, drives the saliency maps closer to the GT. Thanks to the three losses, the model can continuously update the soft association map to obtain high-quality superpixels and better saliency maps.

Table I shows the training steps of the proposed superpixel-wise lacunarity saliency algorithm. The input SAR images consist of the unlabeled data $\mathbf{I}_U$ and labeled data $\mathbf{I}_L$. First, for each image, the encoder extracts the features and the decoder predicts the soft association map $\mathbf{Q}$, which holds the probability that each pixel belongs to all superpixels in this image. Second, given the $\mathbf{Q}$, we can obtain the intensities and locations of all superpixel centers. The equivalent intensity and location of each pixel can be then reconstructed. On this basis, for all input ($\mathbf{I}_U$ and $\mathbf{I}_L$), we can obtain the losses $L^{\mathrm{u}}_{\mathrm{recons}}$ and L_{compact} according (7) and (8). Third, the LSM is calculated based on association map $\mathbf{Q}$. For the labeled input $\mathbf{I}_L$, we can get their saliency loss L_{Sal}. Finally, the total loss L_{total} is used to iteratively update the parameters of superpixel generation network.

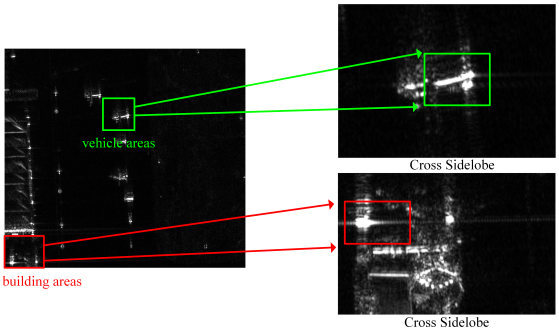

Fig. 8. Cross sidelobes in SAR images.

D. Postprocessing on LSMs

Although the saliency map obtained above has removed most of the background clutters, some man-made objects may remain high saliency values in the saliency maps. Hence, a postprocessing step is introduced to further remove the false alarms.

1) Sidelobe Suppression: In SAR images, there are always obvious sidelobes centered on the strong scattering points. These sidelobes are caused by small beams accompanying the largest radiation wave, i.e., the main lobe. As shown in Fig. 8, such sidelobes often appear as brighter crosses in SAR images. During the calculation of lacunarity saliency, these sidelobes are easily mistaken as strong echoes from vehicle targets and are given high saliency, thus resulting in false alarms in the detection process.

Some scholars proposed to use the "cross"-shaped templates parallel to the azimuthal or range directions to convolve with SAR images to remove the sidelobes. However, in many SAR images, the sidelobes rotate at a certain angle. At this time, there is a certain angle between their azimuthal sidelobes and azimuthal axis. Only the template parallel to the azimuthal or range directions cannot adequately remove the sidelobes.

Therefore, this article proposes a convolutional template set for sidelobes suppression in SAR images, which contains templates of two different sizes (i.e., 55 × 55 and 25 × 25) combined with two different directions (i.e., 0 and 45°). These templates are used to convolve with the LSMs, respectively. Then, these templates are combined in a weighted average manner to form the convolved saliency maps. The result after convolution operation can be obtained by the following calculation:

$$\mathrm{Comb} = \frac{1}{\mathrm{Te}} \sum_{\mathrm{temp}=1}^{\mathrm{Te}} \mathrm{Sal}^{\mathrm{temp}} \tag{17}$$

where Te is the number of templates, which is four in this article. $\mathrm{Sal}^{\mathrm{temp}}$ is the convolution result corresponding to the tempth template.

Then, Comb as input is again convolved with the template according to (17). It will further enhance the targets regions and suppress the sidelobes. This process will be carried out iteratively until the histogram similarity sim_i between the saliency maps before and after convolution remains essentially unchanged. In this article, the condition for the iteration to be

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terminated is set to $\Delta_i \leq 1\%$, which is defined as

$$\Delta_i = |\text{sim}_i - \text{sim}_{i+1}|/\text{sim}_i. \tag{18}$$

When $\Delta_i \leq 1\%$, that is, the similarity is basically constant. In this case, the absolute difference between the two similarities is small enough and the iterative process can be terminated.

2) Morphological Processing: Considering that the size prior information of targets is known, we introduce the reconstructed open operations [35] to suppress these clutters as follows:

$$\text{Mor}(\text{dir}, \Omega) = \text{open}^{re}(\text{dir}, \Omega, \text{Comb}) \tag{19}$$

where $\text{open}^{re}(\cdot)$ is the reconstructed open filter. dir and Ω are the direction and length of the linear structure element of the filter. Comb is the LSM after sidelobe suppression. $\text{Mor}(\text{dir}, \Omega)$ is the filtering result. In principle, the objects in the input image whose size is greater than or equal to Ω along the dir direction will be retained and the bright objects with the lengths smaller than Ω along the direction dir will removed.

Let $\Omega_{\max}$ and $\Omega_{\min}$ represent the maximum and minimum length of the targets to be detected in the SAR images, respectively. In $\text{Mor}(\text{dir}, \Omega_{\max})$, all clutters or targets smaller than $\Omega_{\max}$ in the direction of dir will be removed. Similarly, in the results of $\text{Mor}(\text{dir}, \Omega_{\min})$, all saliency areas smaller than $\Omega_{\min}$ along the direction dir will be removed. Therefore, the absolute difference between these two filtering results, i.e., DMor(dir), can preserve the bright objects whose lengths are between $\Omega_{\min}$ and $\Omega_{\max}$ while removing the other bright objects along the direction dir. DMor(dir) is calculated as

$$\text{DMor}(\text{dir}) = |\text{Mor}(\text{dir}, \Omega_{\max}) - \text{Mor}(\text{dir}, \Omega_{\min})|. \tag{20}$$

Only the clutter areas larger than the minimum target size and smaller than the maximum target size will be maintained in the filtered results.

According to the suggestion of [35], dir $\in \{0°, 45°, 90°, 135\}$. Since the targets may locate along any directions in an SAR image, the final morphological processing results are the combination of the DMor with different directions. It can make the morphological processing results more robust. Note that $\Omega_{\max}$ and $\Omega_{\min}$ should be set according to both the lengths of the targets and the resolutions of SAR images. In the experimental section, we will give details of how to set these two parameters.

On the other hand, we utilize the prior information about the shapes of targets to further reduce the false alarms. Taking vehicle targets for example, the shapes of the vehicle targets have higher similarity with the rectangle, while the shapes of the clutters are various. Therefore, the rectangularity is used in this article to indicate how much a region's shape resembles a rectangle. Rectangularity is defined as

$$\text{recsim}_i = \frac{\text{Area}_i}{\text{Area}_{\text{rec}}} \quad (i \in H_n) \tag{21}$$

where $i \in H_n$ indicates a high saliency region, Area_i is its area (number of pixels), and Area_{rec} is the area of the minimum external rectangle. The rectangularity indicates the area ratio of between the region and the minimum external rectangle. The higher the rectangularity, the closer the shape is to the rectangle, and the more likely it is a target region. And potential regions whose rectangularity is smaller than the threshold will be considered as clusters and removed from the detection results.

IV. Experimental Results and Analysis

In order to verify the method proposed in this article, we perform experiments on real SAR images with different sizes and complexity. First, the data and related parameter settings used in the experiments are introduced in detail. Then, we qualitatively and quantitatively analyze the proposed method's results and compare them with other SAR saliency methods and some classical detection algorithms.

A. Experimental Data and Evaluation Indexes

1) Experimental Data: In order to measure the effectiveness of the method in complex SAR data, this article tests on four real SAR images in complex scenes. It should be noted that the vehicles are recognized as targets in our experiments and the other parts are regarded as clutters. This is first because these movable objects change their positions frequently and need to be redetected every short period of time. Second, it is more difficult to detect movable objects such as vehicles due to their smaller areas. Taking vehicles as the target of interest can better verify the effectiveness of the proposed saliency method. Therefore, only the vehicles are labeled in GT instead of buildings or other man-made objects.

In the saliency detection experiments, the testing dataset is composed of four images: Image I is obtained by the Institute of Electronics, Chinese Academy of Sciences at Yaocheng Airport in Taiyuan, Shanxi, China, in July 2019. Its resolution is 0.15×0.15 m and the size is 5000×5000 pixels. The image contains neatly arranged vehicles and surrounding towers, hangars, sports fields, and other buildings. The original image and the GT are shown in Fig. 9(a), i.e., the first column. Image II, Image III, and Image IV are from the typical miniSAR dataset [43]. The dataset was acquired by Sandia National Laboratories, Livermore, CA, USA, with a resolution of 0.1×0.1 m. The scene is very complex, including closely arranged vehicles, grasslands, roads, and buildings with high scattering intensity and many trees. The size of Image II is 2507×1625 pixels, and the sizes of Image III and Image IV are both 2500×2500 pixels. The original images of these three images are shown in the first row of Fig. 9(b)–(d), and the corresponding GTs are shown in the second row of Fig. 9(b)–(d). Note that the superpixel generation network requires that the input size is 512×512, so all large images are all need to be cropped into this size.

While the training dataset includes 993 labeled SAR images with a size of 512×512. They are composited according to the data preparation strategy in [50]. Specifically, the samples are formed by blending the SAR images of background and object. The object images are selected from the open-source experimental database MSTAR according to the visual analysis. The background surface datasets consist of X-band SAR images with a spatial resolution of 0.3×0.3 m from the open

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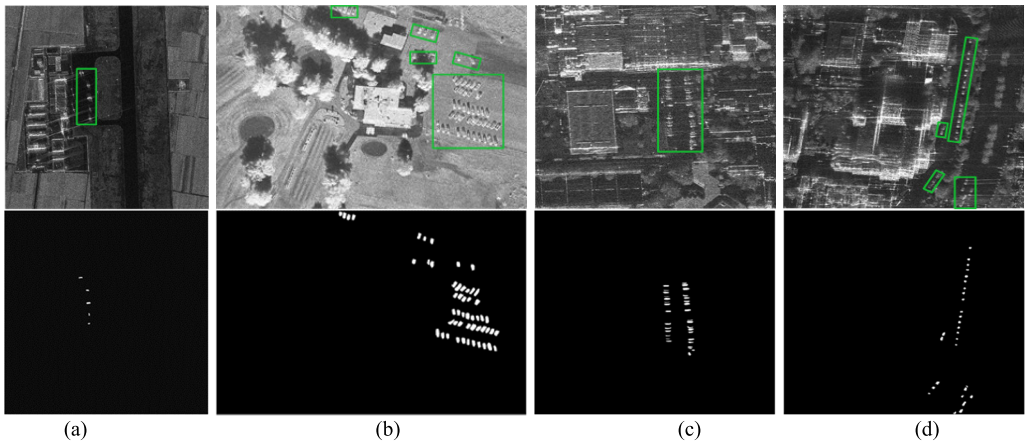

Fig. 9. SAR images in testing dataset and corresponding GT. (a) SAR Image I. (b) SAR Image II. (c) SAR Image III. (d) SAR Image IV.

ICEYE database [50]. The background contains grass, trees, and a small number of man-made objects such as roads and buildings.

When training the proposed superpixel-wise lacunarity saliency method, the abovementioned training dataset (labeled data) and testing dataset (unlabeled data) are input into the soft superpixel generation network together. Note that the training dataset refers to the labeled data $\mathbf{I}_L$ in Table I and the testing dataset, namely, is the unlabeled data $\mathbf{I}_U$ in Table I. Then, three kinds of loss can be calculated using the predicted superpixel generation results and LSMs. Based on this, the superpixel generation network can progressively refine the details of superpixels step by step.

2) Comparative Methods: Six saliency algorithms are selected as the comparison methods to verify the performance of the proposed method, such as background and foreground seed (BFS) [44], multiscale CFAR [45], luminance contrast (LC) [16], regional stability saliency (RSS) [46], Bayesian morphological saliency (BMS) [47], and deep hierarchical saliency (DHS) [48]. Among them, BFS, LC, and BMS are all saliency methods. BFS acquires the saliency map by selecting seed points from the foreground and background separately. It also designed a Gaussian falloff operation to refine the saliency maps. LC is a classic and widely used saliency method, which constructs the saliency maps by using the global contrast of images and has a good performance in SAR images under simple scenes. BMS generates the saliency maps based on Bayesian model. Note that it has a similar postprocessing process with this article, such as open operation. Multiscale CFAR and RSS are both widely used target detection methods. In multiscale CFAR, a multiscale detector is designed to detect maritime targets based on the assumption that clutters obey the generalized gamma distribution. RSS extracts the ROIs by combining the local contrast-based saliency maps and the Boolean maps of stationary areas. DHS is an end-to-end deep hierarchical saliency network based on convolutional neural networks for detecting salient objects. DHS first makes a coarse global prediction by learning various global structured saliency cues. Then, a hierarchical recurrent convolutional neural network is adopted to further hierarchically and progressively refine the details of saliency maps. In our experiments, DHS is also trained using the training dataset including 993 composited SAR images and corresponding GT saliency masks. Then, we test DHS on the testing dataset composed of four images. The network structure and hyperparameters of the DHS remain consistent with [48].

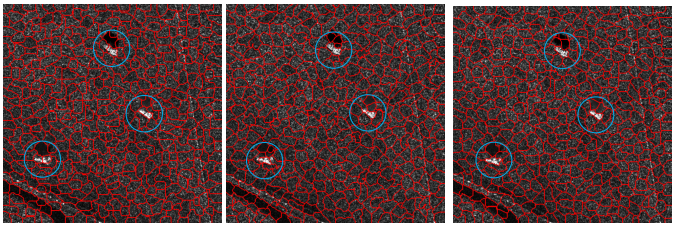

Fig. 10. Change of superpixel generation results under different numbers of epochs: (a)–(c) superpixels from our method when epoch is 10, 20, and 50, respectively. The three vehicles are marked by blue circles.

3) Evaluation Indexes: In the evaluation of the experimental results, in addition to the qualitative analysis of the results from the perspective of human visual effects, this article also introduces three indexes: P–R curve, F_1, and correlation coefficient [49]. The P–R curve is a relationship curve drawn by the recall rate (Recall) as the horizontal coordinate and the precision rate (Precision) as the vertical coordinate. The closer the curve is to the upper right, the better the detection effect of the algorithm.

F_1 is defined as

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{22}$$

The correlation coefficient ρ between saliency map and corresponding GT is also used as an evaluation index in this article

$$\rho = \frac{\sum_{i,j} \left[\left(f(i,j) - \mu_f\right)\right] \times \left[(\text{GT}(i,j) - \mu_{\text{GT}})\right]}{\sqrt{\sigma_f^2 \times \sigma_{\text{GT}}^2}} \tag{23}$$

where $f(i, j)$ and $\text{GT}(i, j)$ are the saliency map and GT, respectively, μ_f and μ_{GT} are the mean of the two maps, respectively, and σ_f^2 and σ_{GT}^2 are the variance of the two

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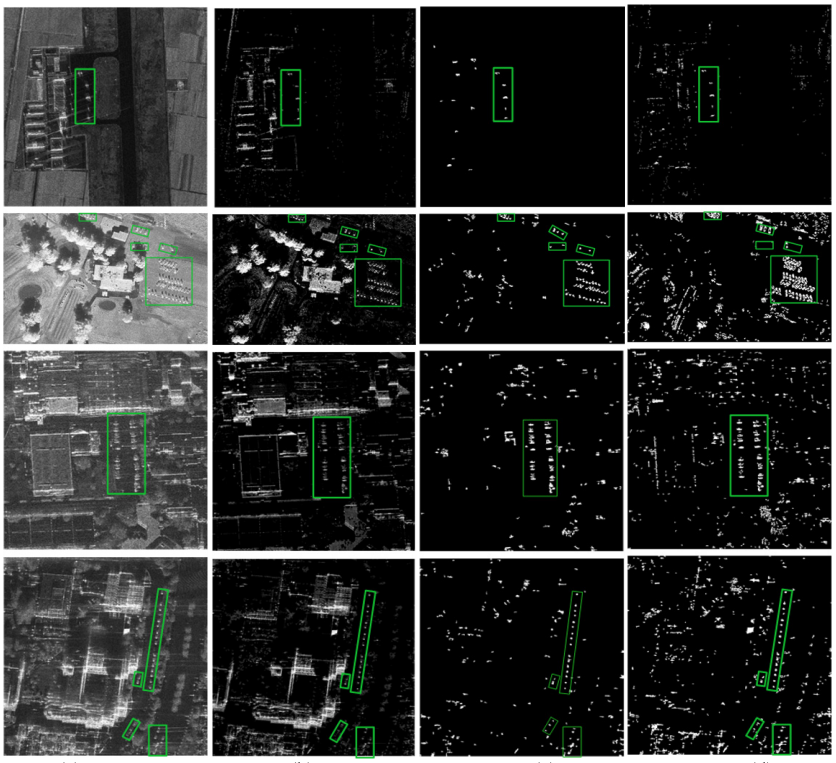

Fig. 11. Results of different steps of the proposed method on four SAR images: (a) original SAR images; (b) LSMs; (c) postprocessing on the LSMs; and (d) postprocessing on the original SAR images. The green boxes are the target areas to be detected.

maps, respectively. The larger the F_1 and ρ, the better the detection effect of the saliency maps.

B. Parameter Settings and Training Process

In the superpixel-wise lacunarity saliency calculation stage, we use a grid of size 16×16 as the initial superpixels. The box size is set as 7×7. The model is optimized using Adam with learning rate $= 0.0001$, training epoch $= 100$, and batch size $= 100$. According to the suggestions in [38] and later adjustment, we set $\alpha = 0.1$ and $\beta = 1$ in experiment. In addition, these experiments are conducted on Intel[1] Core i5-3470 CPU at 3.20-GHz computer.

In the postprocessing stage, $\Omega_{\max}$ and $\Omega_{\min}$ in (19) and (20) are determined jointly according to the length of the real targets and the resolution of the SAR images. The maximum and minimum lengths of real vehicle targets are about 5.0 and 1.0 m, respectively. Image I has resolutions of 0.15×0.15 m, and thus $\Omega_{\max}$ and $\Omega_{\min}$ are set to 33 and 7 pixels, respectively. Image II, Image III, and Image IV are all from miniSAR and have resolutions of 0.1×0.1 m. Hence, $\Omega_{\max}$ and $\Omega_{\min}$ are set to 50 and 10 pixels in the following experiments. The threshold of rectangularity is set as 0.70.

The main advantage of our method is that the shapes of superpixels can be updated to better fit the boundaries in SAR images during the training process. Fig. 10 shows the superpixel generation results of the network under various epochs for an SAR image in training dataset. There are three vehicle targets parked in the grass. Fig. 10(a)–(c) shows the superpixel generation results when the epoch is 10, 20, and 50, respectively. When the epoch is 10 and 20, some superpixel shapes don't fit the shapes of targets well. Some target pixels are clustered into the same superpixels with the surrounding background pixels. And when epoch is 50, superpixels can better capture the targets' shapes. In other words, all pixels in a vehicle region are assigned to the same superpixel. It can be seen that the superpixel generation performance is continuously optimized during the training process.

C. Saliency Maps and Detection Results

1) Saliency Maps and Detection Results of the Proposed Method: This section aims to verify the effectiveness of each step of the proposed method, i.e., lacunarity saliency calculation and postprocessing. Fig. 11 shows the results of different steps of the proposed method on four SAR images. Fig. 11(a)–(d) shows the original SAR images I–IV, the LSMs, the postprocessing results on the LSMs, and the postprocessing results without LSMs. The green boxes are the target areas to be detected.

It can be seen from Fig. 11(b) that the superpixel-wise LSMs can roughly distinguish the target regions and background areas. While removing most of the natural terrains, the target areas are well-preserved. The following postprocessing on the LSMs [Fig. 11(c)] can effectively remove the clutters whose sizes and shapes are different from the vehicle targets while preserve the target areas well. Hence, the final results have few false alarms and high detection rate. In contrast, the

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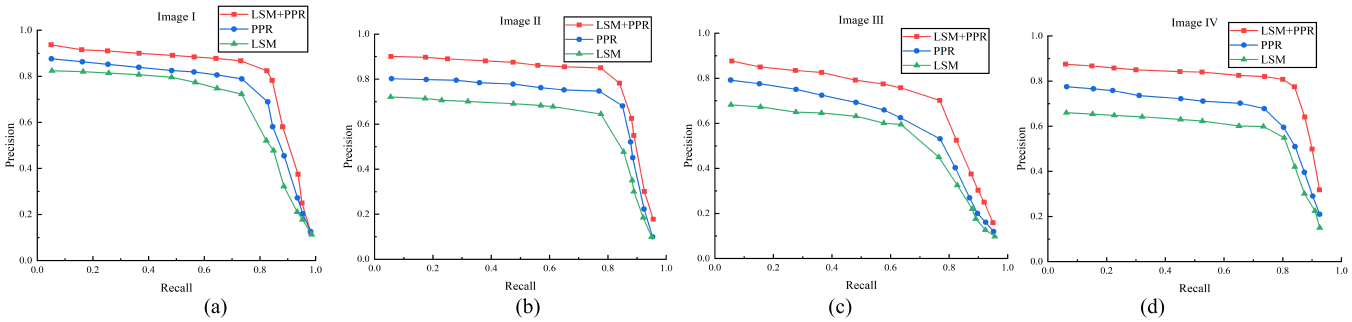

Fig. 12. Precision-recall (PR) curves at different steps of the proposed method: (a) Image I; (b) Image II; (c) Image III; and (d) Image IV.

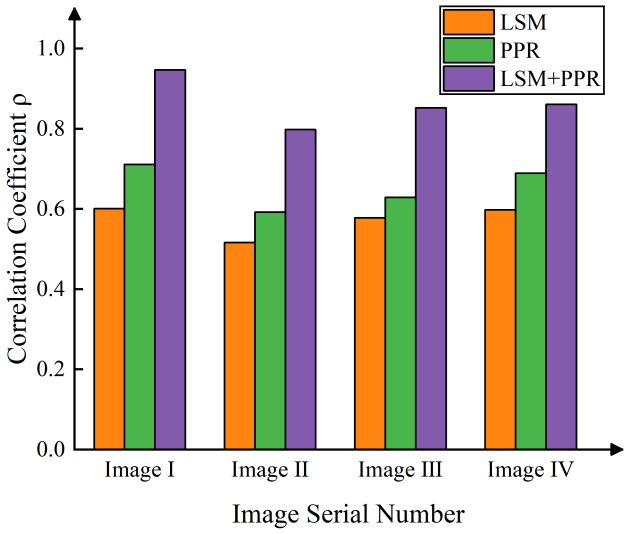

Fig. 13. Correlation coefficients of different steps on four SAR images.

postprocessing results without the LSMs [Fig. 11(d)] can only remove the false alarms with large areas, such as buildings, but those false alarms with similar sizes to the targets cannot be effectively recognized.

In order to quantitatively verify the effectiveness of each step, we, respectively, compare the P–R curves, correlation coefficients, and F_1 of LSMs, postprocessing without LSMs (PPR) and postprocessing on the LSMs (LSM + PPR). Fig. 12(a)–(d) gives the superpixel level P–R curves in turn on Images I–IV. The comparison of the proposed method (i.e., LSM + PPR) and postprocessing without LSMs (i.e., PPR) shows that the lacunarity saliency is essential for vehicle target detection task. Specifically, with the same recall rate, LSM + PPR can improve the precision by about 0.12–0.18, which is consistent with Fig. 11. Compared with the results of LSM, the P–R curves of LSM + PPR are closer to the upper right, thus proving that postprocessing can further improve the detection performance.

Fig. 13 shows the correlation coefficients of different steps on four SAR images. Compared LSM with LSM + PPR, it can be seen that the introducing of postprocessing on LSMs improves the correlation coefficients on average by about 0.25. Comparison of PPR and LSM + PPR reveals that lacunarity saliency can increase the correlation coefficient on average by 0.21. Moreover, the correlation coefficients of LSM + PPR are all more than 0.80. In the relatively simple scenes of Image I, it reaches above 0.90, which verifies the effectiveness of each step of our method.

TABLE II
F_1 AT DIFFERENT STEPS OF THE PROPOSED METHOD ON FOUR IMAGES

F_1	Image I	Image II	Image III	Image IV
LSM	0.70	0.67	0.60	0.61
PPR	0.73	0.71	0.65	0.71
LSM+PPR	**0.83**	**0.83**	**0.78**	**0.80**

The F_1 of different steps on Images I–IV is shown in Table II, and the threshold is determined by the iterative selection method. It can be seen that the LSM and PPR can both improve F_1 to a certain extent. All final F_1 reaches around 0.80 on four images. Especially in the complex scenes (such as Image III, IV), the results of LSM are not satisfactory. After introducing PPR, the F_1 values of the results are significantly improved. Overall, the final results are relatively stable and have little fluctuation in images with different scene complexity.

2) Comparison of Saliency Maps With Different Methods: This section compares and analyzes the saliency maps of the proposed method with other saliency methods and classical detection algorithms, such as BFS, multiscale CFAR, LC, RSS, BMS, and DHS. Among them, BFS, LC, BMS, and DHS are all saliency detection methods. Therefore, Fig. 14 shows the comparison of the saliency maps generated by the above methods with that of the proposed method. The other two methods, multiscale CFAR and RSS, are classical target detection methods, so their detection results are directly displayed. It should be noted that this section mainly focuses on analyzing the performance of LSMs and does not pay attention to the subsequent processing. The green boxes are the target regions and the red boxes are the missed parts.

Fig. 14(a) shows the original SAR images. Fig. 14(b) shows the saliency maps of BFS method. It can be seen that many targets in SAR images are missed, and the shapes and structures of some targets are incomplete. Fig. 14(c) shows that the targets are basically detected by the multiscale CFAR, and only limited number of targets is missed. However, it cannot keep vehicles' shapes and structures completely.

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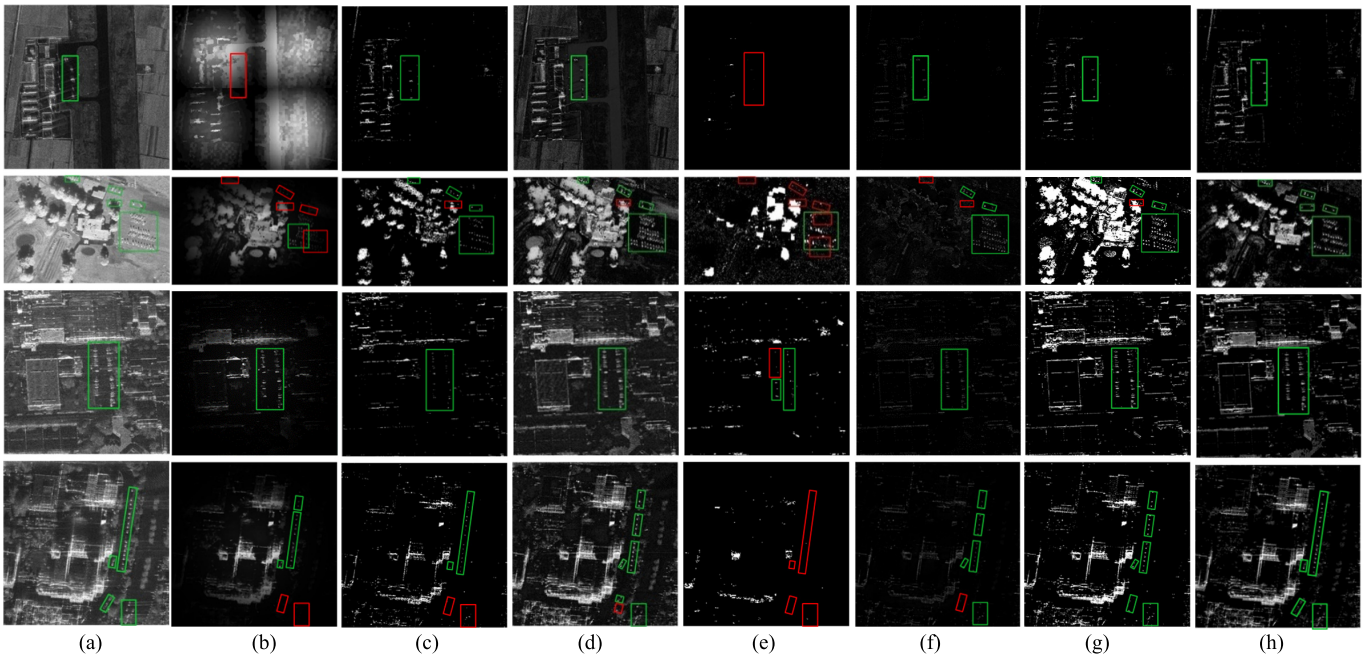

Fig. 14. Comparison of saliency maps of different methods on four SAR images: (a) original SAR images; (b) BFS; (c) multiscale CFAR; (d) LC; (e) RSS; (f) BMS; (g) DHS; and (h) proposed method. The green boxes are the target regions and the red boxes are the missed targets.

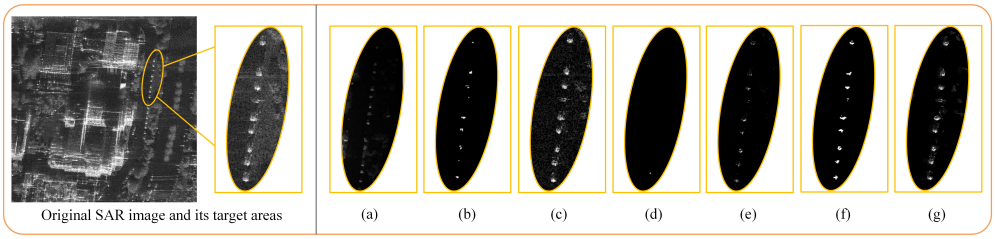

Fig. 15. Local targets areas on the saliency maps of different methods: (a) BFS; (b) multiscale CFAR; (c) LC; (d) RSS; (e) BMS; (f) DHS; and (g) proposed method.

Fig. 14(d) shows that LC can basically get the target areas, but background areas are hardly suppressed at all compared with other methods. Fig. 14(e) shows the results obtained by the RSS. This method cannot well suppress the building areas which generally have strong backscattering coefficients and more targets are also missed. As shown in Fig. 14(f), the saliency maps of BMS show that it effectively suppresses the background clutters, and only a few targets are missed. As can be seen from Fig. 14(g), DHS is easily distracted by local salient patterns and are not able to highlight salient objects in cluttered backgrounds (Images II–III). Our saliency maps shown in Fig. 14(h) not only have the least misses, but also can keep vehicles' shapes and structures more completely.

Fig. 15 shows the enlarged target areas on the saliency maps of different methods. It can be seen that, only LC [Fig. 15(c)] and the proposed method [Fig. 15(f)] can retain the shapes of the targets well. Other methods have different degrees of miss detection. Our method is superior to LC in suppressing background clutters. In conclusion, consistent with the above analysis, the proposed method can not only better preserve the structures and shapes of the targets, but also better suppress most of the surrounding background clutters, leading to the least false alarms.

In order to more accurately evaluate the performance of our saliency maps, we quantitatively compare the P–R curves, correlation coefficients, and F_1 of different methods' saliency maps. Fig. 16(a)–(d) shows the P–R curves of different methods on Images I–IV at superpixel level. It can be seen that in the experimental results on four images, the proposed method's P–R curves are closer to the upper right and completely surround the curves of the other methods. At the same recall, the proposed method's precision is far higher than that of other methods. This shows that our method outperforms the other methods by a large margin. In addition, the performance of our method for images with different scene complexity is most stable, indicating that the method has the better feasibility. DHS's performance is second only to our method. The hierarchical refinement scheme proposed in DHS helps to progressively eliminate false alarms. Its curves are higher than other five comparative methods from the beginning to the end on all four SAR images.

Fig. 17 shows the comparison of the correlation coefficients between the binary maps obtained by different methods'

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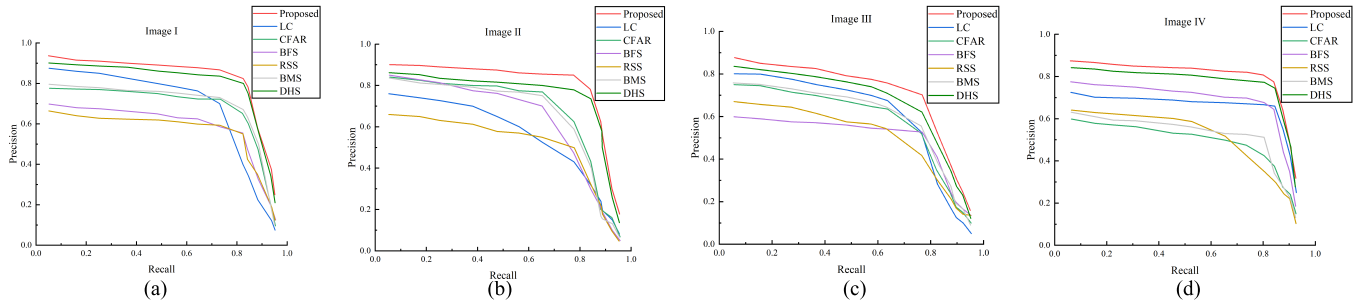

Fig. 16. P–R curves of different methods' saliency maps on four SAR images: (a) Image I; (b) Image II; (c) Image III; and (d) Image IV.

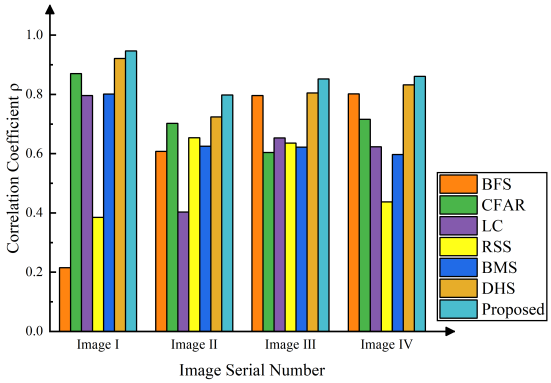

Fig. 17. Correlation coefficients of different methods' saliency maps on four SAR images.

TABLE III
F_1 OF DIFFERENT METHODS ON FOUR IMAGES

F_1	BFS	CFAR	LC	RSS	BMS	DHS	Proposed
I	0.61	0.72	0.71	0.62	0.70	0.80	**0.83**
II	0.69	0.74	0.59	0.46	0.58	0.70	**0.83**
III	0.57	0.62	0.67	0.58	0.61	0.72	**0.78**
IV	0.69	0.51	0.68	0.58	0.60	0.76	**0.80**

saliency maps and corresponding GT on Images I–IV. It can be seen that our method has the highest ρ for all four images, reaching 0.80 on average. The ρ of DHS is slightly lower than the proposed method, while the other algorithms basically do not reach 0.8 on Images II–IV. BFS's ρ on Image I only about 0.23, and our method reaches 0.92 accordingly. It indicates that the proposed algorithm is far superior to other algorithms.

In Table III, the F_1 of different methods for Images I–IV is shown, and the threshold is determined by the iterative selection method. It can be seen that the proposed method obtains the highest F_1 for each image, which close to 0.83 when the scene is relatively simple. In images with complex scenes (such as Images II–IV), the F_1 of the proposed method can reach about 0.78. In contrast, the F_1 of other methods are relatively low, and even fails to reach 0.70. This indicates the proposed method has more stable performance on different imaging scenes.

3) Comparison of Detection Results With Different Methods: This section aims to compare the detection performance for vehicles of different methods. In order to ensure the fairness of the comparison, we perform postprocessing on the saliency maps in Fig. 14 of the six comparison methods in this section. It should be noted that the BMS method itself has a similar postprocessing algorithm, so its own postprocessing is used here. The other methods use the same postprocessing as this article. As shown in Fig. 18, the target areas are labeled by green boxes.

It can be seen intuitively that the number of targets detected by different methods on each image are quite different. Among the seven methods, RSS detects the least number of targets and has poor detection performance even in SAR image with the simplest scenes (such as Image I). BFS also has poor detection performance. Especially, those at the border of the images are easily missed. The vehicles in Image II are arranged tightly, thus greatly increasing the difficulty of detection. For example, in LC's results, a lot of targets are missed in Image II. In addition, the multiscale CFAR detects most targets, but some targets are also missed, especially in complex scenes (such as Images II–IV). BMS detects more targets, but in Image II, some shadows of the vehicles are regarded as targets, which brings a lot of false alarms to the results. DHS can detect most of targets on four images, which is better than the other five unsupervised comparative methods. However, it cannot handle various situations well, including complex backgrounds (Image III) and objects touching image boundaries (Image IV). In contrast, the proposed method can overcome these difficulties well. It can detect all the targets while keeps the less false alarms than multiscale CFAR, LC, BMS, and DHS.

Based on Fig. 18, we can obtain the average detection rate and the number of false alarms of all methods. As shown in Fig. 19, the proposed method has the highest detection rate with an average detection rate of 0.91. While the best of the other six methods (DHS) reaches 0.87, and the rest of the methods are all below 0.82. In terms of the average number of false alarms, ours far less than CFAR, LC, BMS, and DHS on all four images. Although BFS and RSS have fewer false alarms, their detection rates do not exceed 0.5. In summary, the proposed method has great advantages in terms of detection rate and the number of false alarms. It verifies the feasibility and effectiveness of the method for SAR images in complex scenes.

D. Effect of Box Size L and Superpixel Size on Results

In the construction of LSM, the selection of the box size L and superpixel size A has a great influence on the results.

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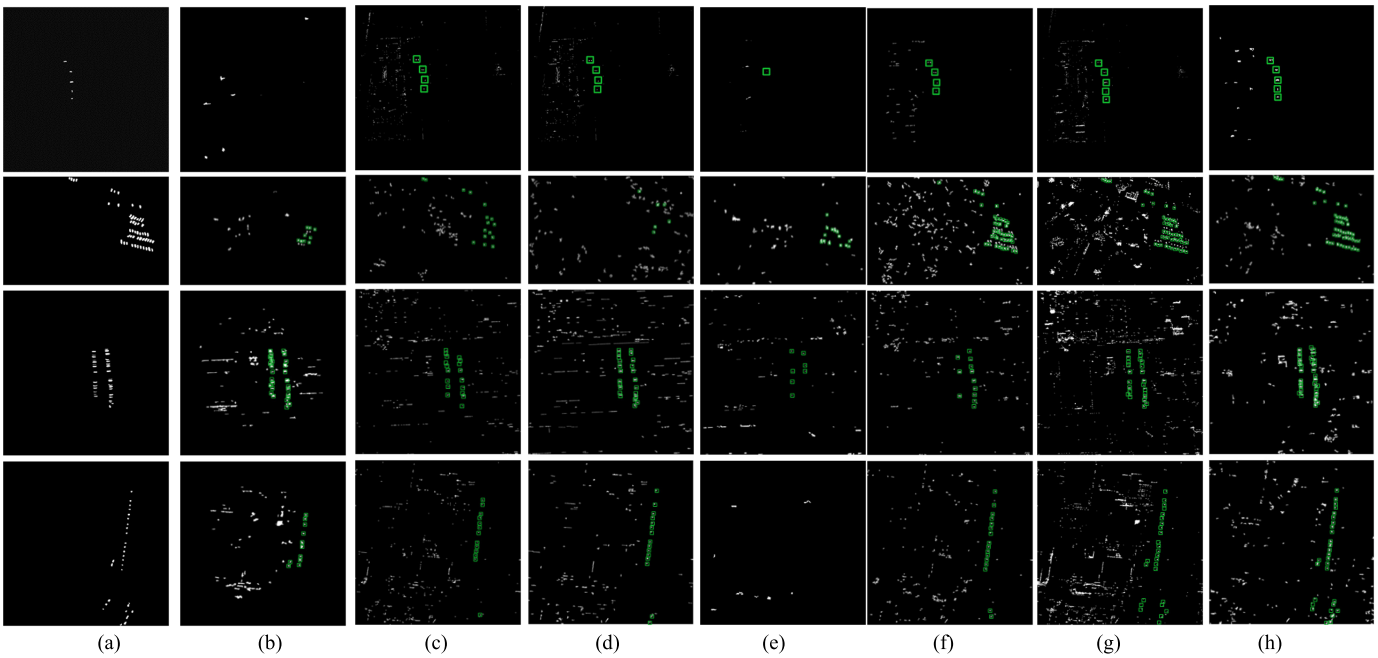

Fig. 18. Comparison of detection results of different methods on four SAR images: (a) GT; (b) BFS; (c) multiscale CFAR; (d) LC; (e) RSS; (f) BMS; (g) DHS; and (h) proposed method. Each green box represents a detected target.

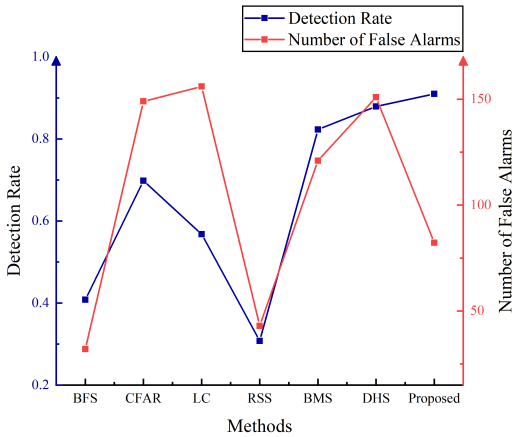

Fig. 19. Detection rate and the number of false alarms of different methods.

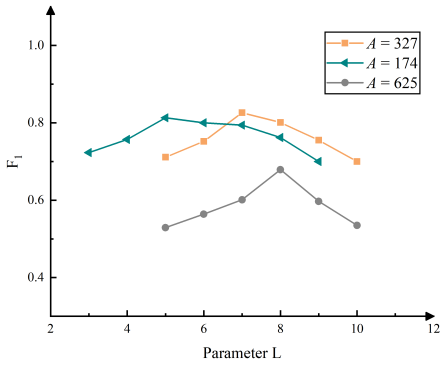

Fig. 20. F_1 under different box sizes L and superpixel sizes A.

When L is too small, there are too few reference points when calculating the mass of each box, easily leading to inaccurate masses of the boxes. When L is too large, there will too few boxes in each superpixel, thus resulting in errors in evaluating the distribution of boxes' masses. Improper L finally will affect the measurement of the regularity of the scattering points in the superpixels. Different superpixel size will also bring different detection results. If superpixel size A is too large, the superpixels cannot well adhere the boundaries in SAR images. Many superpixels may contain pixels from the target and background at the same time. Too small superpixel size will also limit the box size L in the superpixel, thus affecting the final detection results.

In order to study the effect of L and A on the results, we compare the F_1 values of the detection results under different superpixel sizes A and box sizes L. Taking Image II as an example, we set the superpixel size A as 174, 327, and 625, respectively, and let L increase from 3 to 12. The corresponding F_1 values are shown in Fig. 20. From the superpixel size perspective, the method can obtain better detection results when the superpixel size is at 174 and 327, with F_1 reaching about 0.81 in both cases. It is because the small superpixel can better adhere the boundaries of the targets. However, when the superpixel size is 625, the F_1 is at a very low level. In this case, superpixels cannot fit the boundaries well. Therefore, the better the superpixel generation results are, the more it helps saliency detection. In this experiment, a superpixel size of 300 is a desirable value. On the other hand, box size L also greatly affects the performance of the method. It can be seen that when $A = 327$, the best choice of L is 7, and when $A = 174$, the most appropriate L is 5. Under these two kinds of superpixel sizes, the highest F_1 values are both about 0.81. In summary, the suitable range of values for L is from 5 to 9 when the superpixel size is set reasonably.

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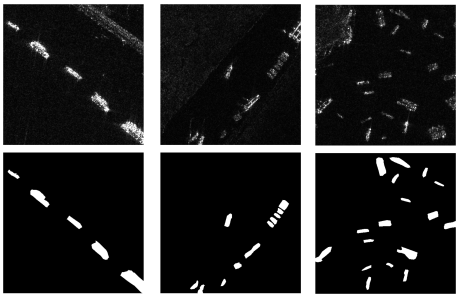

Fig. 21. Three examples from (First row) FUSAR-Ship1.0 and (Second row) corresponding GT.

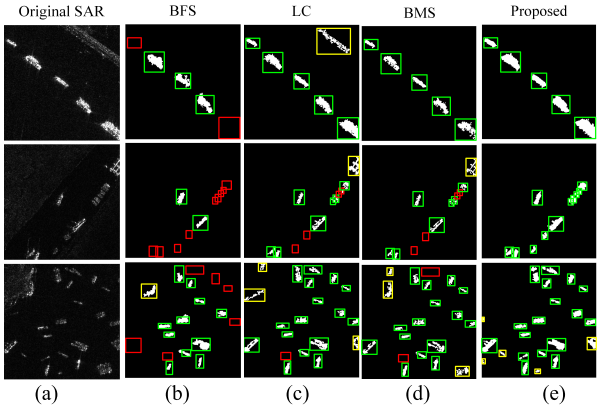

Fig. 22. Comparison of results of different methods on FUSAR-Ship1.0 dataset: (a) original SAR images; (b) BFS; (c) LC; (d) BMS; and (e) proposed method.

E. Saliency Detection Results on Ship Targets

In order to verify that the feasibility of the proposed LSMs, we also conduct an experiment on newly released ship detection dataset in this section, named FUSAR-Ship1.0 [42]. It is constructed on more than 100 Gaofen-3 scenes covering a large variety of sea, land, coast, river, and island scenarios. It includes over 5000 ship image chips with a size of 512 × 512 (Fig. 21). In this experiment, the ships are regarded as the targets and the other parts are clutters. Note that we randomly select half of this dataset as training data and the other half are as testing data. These two parts of data are input into the soft superpixel generation network to train the network as described in Table I. The structure of the network and the hyperparameters are consistent with the vehicle detection experiment.

It should be noted that this section is mainly to verify the feasibility of the proposed LSM. Therefore, we only compared our results with that of three saliency detection methods mentioned in Section IV-A, namely, BFS, LC, and BMS. Table IV gives the average F_1 values of four saliency algorithms on all testing images. Fig. 22 shows the detection results of different methods on three images in FUSAR-Ship1.0. The green, red, and yellow rectangles indicate the correctly detected targets, missed targets, and false alarms, respectively. The proposed method still achieves the best detection results on this dataset. This can prove that the saliency maps proposed in this article have satisfactory feasibility.

TABLE IV
AVERAGE F_1 OF DIFFERENT METHODS ON FUSAR-SHIP1.0

Method	BFS	LC	BMS	Proposed
F_1	0.82	0.85	0.87	0.88

TABLE V
AVERAGE TIME COST OF DIFFERENT METHODS

Method	BFS	CFAR	LC	RSS	BMS	PL	Proposed
Time(s)	713.45	402.45	372.58	597.43	306.63	415.23	**181.32**

F. Discussion on the Efficiency of Different Methods

Considering that this article mainly focuses on the construction of LSMs, so we analyze the computational complexity of the above process in this section. The calculation of LSM mainly comes from two steps. The first is the lacunarity cue of each superpixel. Assuming N superpixels averagely contain ℓ pixels and take the L-neighborhood for each pixel in each superpixel as a box, the time complexity of obtaining $\mathrm{Sal}_{\mathrm{in}}^{m}$ is $O(\ell \cdot N \cdot L)$. Second, the lacunarity cue from the other superpixels, i.e., $\mathrm{Sal}_{\mathrm{out}}^{m}$, requires calculating the lacunarity distance and space distance between each superpixel and the others. The time complexity of this part can be denoted as $O(N^2)$. The total computational complexity is $O(\ell \cdot N \cdot L + N^2)$.

In the pixelwise lacunarity feature extraction algorithm based on differential box dimension, for an image with a size of WH, the sizes of window and box are RR pixels and LL pixels. The time complexity of this algorithm can be denoted as $O(W \cdot H \cdot R^2 \cdot L^2)$. Because the number of superpixels is much smaller than that of pixels, that is, $N \ll W \cdot H$, $O(\ell \cdot N \cdot L + N^2)$ is far less than $O(W \cdot H \cdot R^2 \cdot L^2)$. It shows that the introduction of superpixels greatly reduces the amount of calculation and improves the efficiency.

The average time cost of different methods on four SAR images is shown in Table V. Compared with others, the proposed method has great advantages in time consumption. In addition, we also compared the pixelwise lacunarity saliency method (PL) in Table V. It can be seen that PL's running time is 415.23, far larger than that of the proposed superpixel-wise saliency method, which is consistent with the above analysis.

V. CONCLUSION

Considering that there is a large difference in fluctuation of scattering points between targets and background in SAR images, this article proposes the superpixel-wise lacunarity saliency algorithm. We improve the pixelwise box dimension algorithm to measure the irregularity of scattering intensity of superpixels. Then, this lacunarity saliency method and soft superpixel generation network are combined to construct an end-to-end superpixel-wise saliency prediction network. Superpixels' shapes can be iteratively adjusted according to the saliency maps until the parameters get globally optimal solution. The saliency map experiments on real SAR images show that the proposed LSMs can effectively suppress the

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clutters and highlight the target areas. The saliency detection experiments for vehicle targets prove that the LSMs after postprocessing based on target prior information have fewer false alarms and missed targets.

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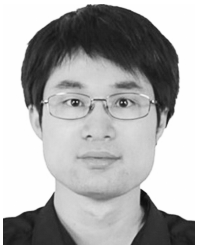

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